Causal Attribution in Causal Chains

Bachelor's Thesis

Submitted in partial fulfillment of the requirements for the degree of Bachelor of Arts in International Studies in Computational Linguistics

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Declaration of Originality

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Place and Date: **Potsdam, 03/02/2026**

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Abstract

This study examines how distal event type and outcome valence framing influence the main causal selections in chains of events. Building on previous research on causal reasoning, it tests whether agency (or human involvement) and outcome framing affect which event is identified as "the main cause" in the chain.

Participants completed an online experiment in which they read short vignettes describing sequences of events leading to positive or negative outcomes. Across trials, the distal event varied in type (natural, non-deliberate, or deliberate), and participants selected the event they considered most responsible for the outcome.

The results showed that deliberate distal events were selected more frequently than natural ones, especially under negative outcome framing. Outcome valence had a modest and conditional influence, strengthening the effect of agency rather than producing a general shift in causal judgments.

These findings suggest that causal selection is shaped not only by structural dependence but also by agency and evaluative context.[1]

[1] All experimental materials, data, and analysis code are available at: `https://github.com/mnadvernyuk/ba-thesis-experiment`

Contents

Introduction

When things happen in our lives, we tend to be quite capable of identifying the reason *why* a certain outcome unfolded one way or another. However, everyday outcomes rarely have a single cause or event that is not related to anything else. Instead, they arise from chains of events where early factors enable later ones. Therefore, in causal cognition, researchers have studied the following question: when multiple events form a chain, which one do we choose as "the cause"?

Findings in causal cognition emphasize dependency and counterfactual reasoning: an event is a candidate for being "the cause" if removing it would prevent an outcome. Nevertheless, research in cognitive science and philosophy of causation shows that people's causal selections are affected by more than just dependency. Factors such as temporal order, causal structure, agency, intentionality, and whether events might violate norms influence our judgment systematically and sometimes even obscurely (Lewis, 1973; Hilton and McClure, 2009; Woodward, 2003; Icard et al., 2017). Strikingly, we often give favour to causes that are agentive, intentional, or norm-violating, even if other "links in the chain"/events were as necessary for the outcome to occur (Hitchcock and Knobe, 2009; Knobe and Fraser, 2008; Icard et al., 2017). As it turns out, when we talk about causal judgments and chains of events, this is intertwined with the question of norm sensitivity and valence: people may shift toward statistically abnormal or atypical factors relative to what is normally observed or expected, but they don't cite the whole chain and simply highlight the abnormalities, typically choosing one event as predominant cause out of the whole sequence (Hilton and McClure, 2009).

Hilton and McClure (2009) directly explored this in their paper using short vignettes of causal chains of events. Participants selected the best "cause" from a sequence of events. Their study showed that people are sensitive to the placement of events in the chain, and that statistical and interpretive principles shape which event is identified as "the cause."

This thesis builds on the solid foundation provided by Hilton's and McClure's research and extends beyond their work by examining causal selection in event chains under two manipulations. Firstly, it covers the type of the event: natural, non-deliberate human, or deliberate human. This manipulation distinguishes natural events from the events caused by humans (accidental or intentional). Previous studies suggest that events involving agency and norm-relevance often become more salient candidate causes than purely physical background conditions (even when other links in the chain are equally necessary for the events to unfold) (Hitchcock and Knobe, 2009; Icard et al., 2017; Henne, 2021).

Secondly, this thesis tries to look into the effects of outcome valence framing. Each scenario appears in both positive- and negative-outcome versions (depending on the participant) to test how participants' evaluations and blame interact with causal judgment. The hypothesis is that negative outcomes might often draw more attention to responsibility, while positive outcomes prompt explanations and elaboration on the circumstances of the event. Harmful outcomes also increase the focus on active interventions rather than background elements (Spranca et al., 1991), and negative outcomes amplify norm sensitivity in causal explanation (Henne, 2021).

The main part of the thesis consists of an online experiment combining a causal-chain selection task (similar to Hilton and McClure) with a randomization scheme and different valences of outcomes (positive vs. negative). Each participant sees each scenario/story, experiences all event types, and receives positive or negative outcomes. These two things let us test whether selections differ by distal type and whether framing produces systematic changes in the choice of explanations/cause attribution.

Background

2.1 Causal explanation in chains of events

Many outcomes result from a series of events: early events set the initial conditions, intermediate events transmit causal influence, and later events produce the outcome. Despite this complexity, people often summarize the sequence with a single highlighted cause. Cognitive science approaches this as causal reasoning with structured representations, selecting key components rather than listing every factor (Gopnik and Schulz, 2007; Lombrozo, 2012).

Causal chains are useful for measuring how people select explanations because every link is part of the process. Choices involve alternatives with different positions and interpretive roles. Earlier links can be enabling conditions, while later links are often viewed as more directly connected to the outcome. Despite that, earlier events might seem more fundamental as they are "setting the scene." Studying causal selection in chains reveals how people weigh proximity and interpretive importance.

In this thesis, each of the six vignettes contains four candidate events explicitly ordered in time:

- **Event A**: precondition / enabling event
- **Event B**: distal event
- **Event C**: proximal event
- **Event D**: immediate event
- **Outcome**: positive or negative

Participants selected which event in the sequence was the best candidate for causing the outcome. This structure closely follows the logic of early causal chain selection paradigms, in which the dependent variable is the distribution of "cause" selections across the chain (Hilton and McClure, 2009).

2.2 The role of intentionality and distal type

Agency, together with intentionality, strongly influences causal attribution. Even when two events are equally necessary, deliberate actions are judged more causally responsible or interpretive than non-agentive events (Hitchcock and Knobe, 2009). Norm violations also shift judgments, as abnormal events become salient explanations (Icard et al., 2017).

This effect is often understood in terms of norms and expectations. Intentional actions are evaluated against social norms and typical behavior, and norm violations stand out as targets for explanation. To add to that, recent evidence suggests that norm sensitivity plays a greater role in cognition, shaping how people interpret causal structure and make responsibility judgments (Henne, 2021). This thesis adopts the abovementioned conclusions by manipulating the distal event so that its causal role remains structurally similar across stories. However, its type varies in perceived agency and control.

In order to test agency effects in an unfolding-chain-of-events format, we manipulated the distal event's type:

Natural: Event B is caused by nature or a non-agentive process (usually something happening without human intervention).

Non-deliberate: Event B is caused by a person, but accidentally or unintentionally.

Deliberate: Event B is caused intentionally by an agent (usually showing an intention that might be harmful or dangerous).

This tests whether people select the event more when it is a deliberate, controllable action. As for the previous findings, deliberate actions should be chosen more often than natural ones (Hilton and McClure, 2009; Hitchcock and Knobe, 2009).

2.3 Outcome valence framing

Apart from the structure of the story and whether an agent took part in it, causal selection can also be influenced by emotional framing. Presenting an outcome as negative rather than positive increases the perceived need for explanation and highlights responsibility (or norm-relevant factors). Classic evidence shows that people perceive harmful outcomes as more strongly caused by active interventions than by omissions, even when the story structures are similar (Spranca et al., 1991).

Outcome framing has also been studied in social evaluation and attribution. Social psychology finds that negative outcomes shift attention to blame-related factors, while positive outcomes reduce the motivation to identify norm violations (Henne, 2021). Framing outcomes as losses or gains changes responsibility and causality attributions in applied settings (Niederdeppe et al., 2010). Even in formal learning, framing interacts with previous beliefs and the evidential value (Pearl, 2009). More generally, it seems that framing has been used not just in formal academic or scientific settings but also in everyday communication, including the media, marketing, and even political messaging.

In this thesis, outcome valence was manipulated by presenting each scenario outcome in either a positive or a negative version, while keeping the chain structure the same. This allowed us to test whether participants were more likely to select certain chain positions under negative framing or under positive framing.

2.4 Summary of research questions and predictions

This thesis addresses the following research questions:

- *Does distal type affect causal selection? (Specifically, if "the cause" event is selected more frequently when the distal event is deliberate than when it is natural or non-deliberate)*
- *Does outcome valence affect causal selection? (Are participants more likely to select the distal event when outcomes are negative or positive)*

Additionally, after seeing the performance of pilot participants (the majority of participants failed the attention check but still completed the experiment), an optional question emerged:

Do attention-check failures correspond to different response patterns?(Do participants who failed the attention check show systematically different selection behavior compared to those who passed it).

The main theoretical prediction for the following research is that participants will select the distal event more frequently when it is deliberate (in line with Hilton and McClure's findings) **than when it is natural.** Overall, this is compatible with norm-based approaches to causal selection and with earlier results suggesting that intentional actions often play a privileged role in causal attribution, even in complex causal events (Hitchcock and Knobe, 2009; Icard et al., 2017; Hilton and McClure, 2009).

Data

3.1 Stimuli and scenarios

The experiment used short text vignettes describing sequences of events leading to an outcome that were accompanied by a neutral illustration (so that the outcome framing is not shown too soon) and a neutral name for the story (the whole set of illustrations and stimuli is available in Appendix A). The stimulus set contains multiple scenarios. In the following thesis, six scenarios were sampled from the stimulus pool. The selected scenario set includes:

- Car incident (similar to Hilton & McClure)
- Apartment building (gas explosion scenario)
- Airport luggage claim
- Theatre performance interruption
- Metro escalator accident
- Cognitive memory test

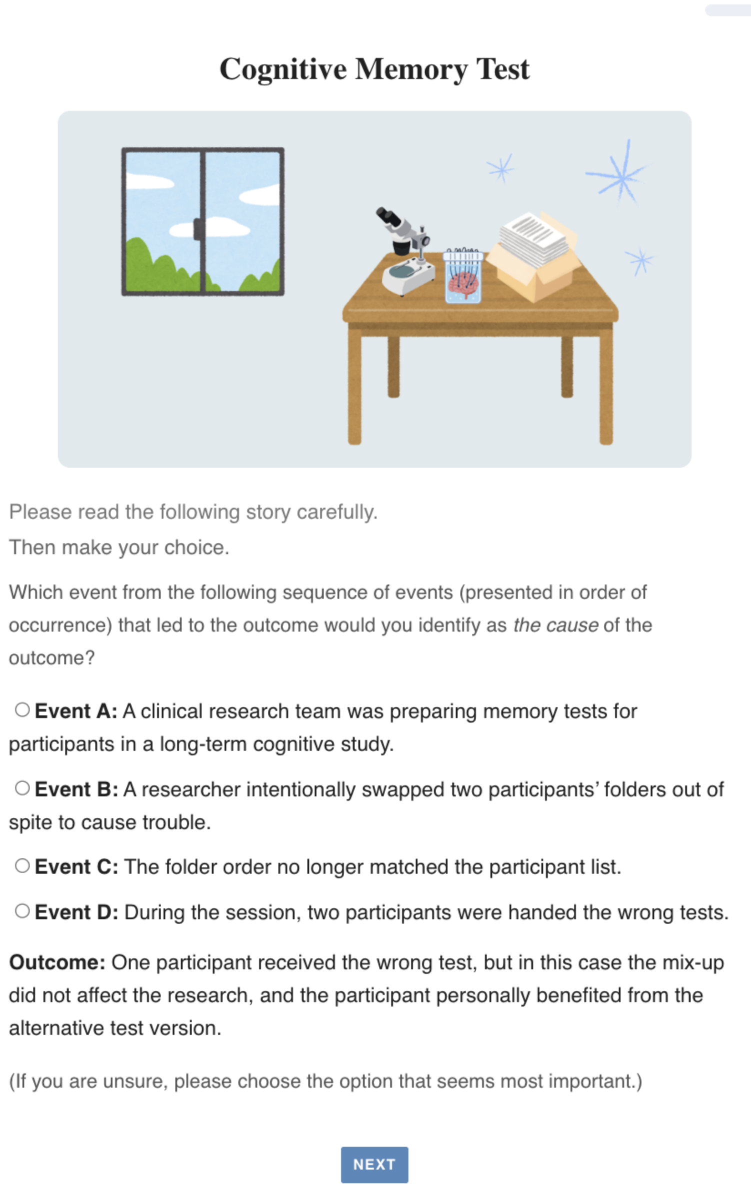

Figure 3.1: Example stimulus screen for the Cognitive Memory Test scenario.

Each scenario existed in multiple versions, allowing the distal event to be assigned to one of three distal types (natural, non-deliberate, deliberate). Additionally, each scenario included two possible outcomes: one positive and one negative, allowing a valence manipulation while keeping the chain structure the same. Each participant saw three positive and three negative outcomes (plus an attention check in the middle of the trials = position 3).

3.2 Experimental randomization constraints

The experiment used a randomization scheme assuring balanced exposure to outcomes:

- Each participant saw each scenario exactly once (six trials of the main experiment).
- Each participant saw each distal type exactly twice (natural, non-deliberate, and deliberate).
- Each participant saw positive outcomes three times and negative outcomes three times.
- The assignment of the scenario $\times$ distal type $\times$ valence was random within each participant.
- The trial order was randomized per participant, with the attention check fixed in the middle.

This confirms that the manipulations were balanced within participants while maintaining a mixed scenario content. The randomization logic and code are available on the experiment's GitHub page.[1]

3.3 Participants and data sources

Data was collected in three phases:

- **Pilot sample**: participants recruited via personal networks for early testing and feedback ($N = 17$).
- **Prolific piloting sample**: participants recruited via the Prolific platform for the pilot part of the study ($N = 15$).
- **Prolific main sample**: participants recruited via the Prolific platform for the main part of the study ($N = 52$).

All three datasets were exported from the experiment server as CSV files, each containing one row per trial, including trial metadata, participant ID, and the chosen event. During analysis, the identifiers were transformed into a single `pid` to ensure consistency during multilevel modeling. After preprocessing, the combined dataset included 82 unique participants (in the ALL dataset, some participants must have submitted the study twice).

[1] `https://github.com/mnadvernyuk/ba-thesis-experiment`

3.4 Attention check and exclusion rule

Each participant completed one attention-check trial inserted at a fixed position in the sequence (third place out of seven). The check required selecting a specific event indicated in the trial and implied by the structure. According to their performance, participants were classified as follows:

- **Pass**: selected the correct event in the attention-check trial.
- **Fail**: selected a different event.

As it turned out, a significant number of participants in the pilot version failed to choose the correct option ($N = 12$ out of 16, compared to $N = 3$ out of 15 in the Prolific pilot sample). Therefore, the analysis had to be slightly adjusted, and two versions were reported to assess whether there was any correlation between the initial hypotheses and attention-check performance:

- **ALL dataset**: includes all participants regardless of attention-check performance.
- **STRICT dataset**: excludes attention-check failures.

This allowed assessment of whether inattentive responding changed the main effects, either slightly or substantially.

Methods

4.1 Task procedure

Participants completed an online causal selection task created using Magpie. They were shown a series of independent short stories describing sequences of four events followed by an outcome. On each trial, participants chose which event was the best candidate for causing the outcome. Responses were made via radio buttons labeled Event A–D.

To avoid confusion, the instructions emphasized that each story should be judged independently and that the events should be understood as a sequence leading to the outcome. Each trial had to be completed (i.e., a choice had to be made) before clicking the "Next" button to proceed to the next story.

At the end of the experiment, participants were asked optional demographic and background questions (age, gender, education, native languages, and comments). These data were recorded but were not required for submission.

4.2 Design and variables

The experiment implemented a within-participant design with the following variables.

Independent variables

- **Distal type** (three levels):
  - natural,
  - non-deliberate,
  - deliberate.
- **Valence** (two levels):
  - positive outcome,

  - negative outcome.

Each participant saw all levels of the distal-type factor (twice each) and both valence levels (three times each).

Dependent variable

The main dependent variable followed the analysis approach of Hilton and McClure (2009):

- **Distal selection**: whether the participant selected Event B,
  - coded as 1 if choice = B,
  - coded as 0 otherwise.

4.3 Analysis approach

The analysis used a Bayesian multilevel logistic regression model at the trial level, with distal selection (choice of Event B) as the outcome. Distal type, valence, and their interaction were included as fixed effects. Additionally, the model included random intercepts for participants (`pid`) and scenarios (`scenario_id`).

Posterior predictions were reported as estimated probabilities of selecting Event B for each distal-type × valence condition, together with 95% highest density intervals (HDIs). These predictions were computed at the population level using only the fixed effects, and the resulting estimates reflected tendencies across participants and scenarios rather than idiosyncratic individual or scenario-specific effects.

The following two contrasts were summarised from the posterior distribution:

- **Deliberate vs. natural** (pooled over valence),
- **Deliberate vs. natural** (restricted to negative outcomes).

For each contrast, results included the posterior mean difference, a 95% HDI, and the posterior probability that the difference was greater than zero.

Results

5.1 Descriptive pattern

As an illustration, distal selection varied across distal type and outcome valence. In the ALL dataset ($N = 82$ participants, 504 trials), the raw distal-selection rates (i.e., the proportions of trials in which Event B was selected) were the following:

- **Deliberate**: negative 0.519 ($n = 77$), positive 0.462 ($n = 91$),
- **Natural**: negative 0.278 ($n = 90$), positive 0.333 ($n = 78$),
- **Non-deliberate**: negative 0.541 ($n = 85$), positive 0.398 ($n = 83$).

The clearest descriptive separation occurred under negative outcomes, where both human-caused distal events (deliberate and non-deliberate) were chosen more often than natural distal events.

In the STRICT dataset ($N = 54$ participants, 330 trials), the raw distal-selection rates were:

- **Deliberate**: negative 0.582 ($n = 55$), positive 0.545 ($n = 55$),
- **Natural**: negative 0.327 ($n = 52$), positive 0.328 ($n = 58$),
- **Non-deliberate**: negative 0.638 ($n = 58$), positive 0.500 ($n = 52$).

Here, both non-deliberate and deliberate distal events were selected more often under negative outcomes than natural distal events, and the overall selection rates were higher than in the ALL dataset.

5.2 Model-based estimates

Because the hierarchical model pooled information across participants and scenarios, the posterior predicted probabilities differ slightly from the raw proportions and include uncertainty intervals.

For the ALL dataset, posterior predicted probabilities of choosing the distal event (Event B) were highest for deliberate and non-deliberate events and lowest for natural events across both outcome valences. In particular, deliberate and non-deliberate distal events showed higher predicted selection probabilities under negative outcomes than under positive outcomes, whereas natural events showed consistently low probabilities with little sensitivity to valence.

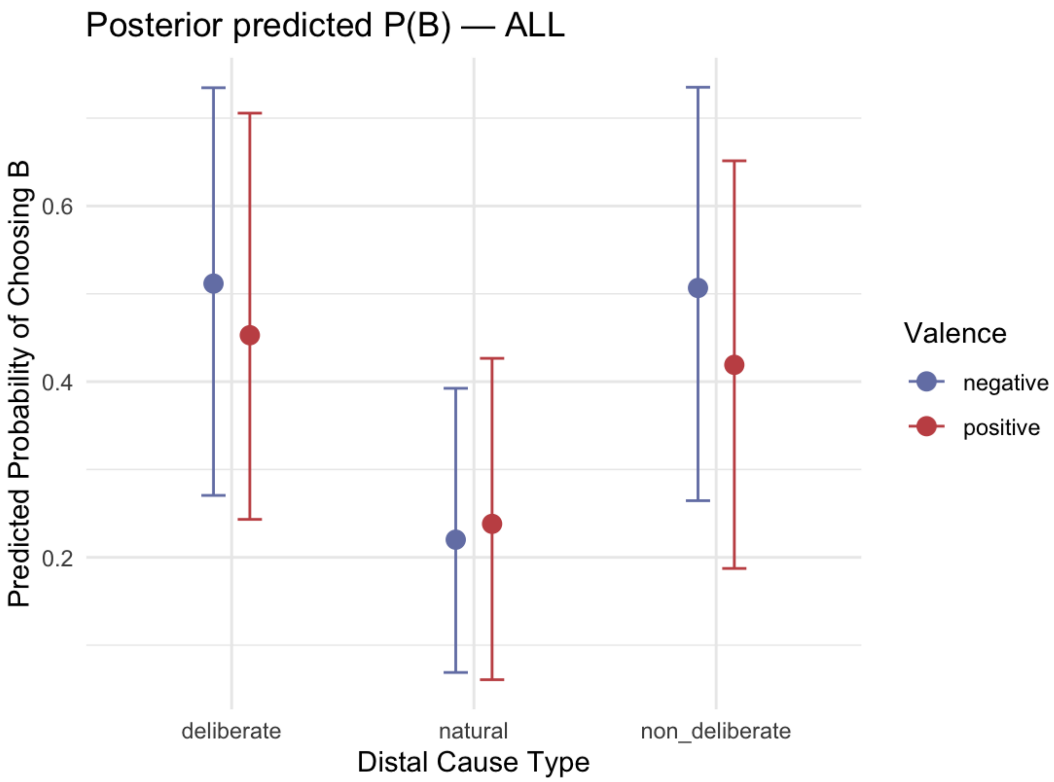

Figure 5.1: Posterior predicted probabilities of selecting the distal event (Event B) in the ALL dataset by distal cause type and outcome valence. Error bars represent 95% highest density intervals (HDIs).

In the STRICT dataset, which excludes participants who failed the attention check, overall predicted probabilities were higher and uncertainty intervals were wider due to the smaller sample size. Nevertheless, the same general pattern was observed: deliberate and non-deliberate distal events were more likely to be selected than natural events, especially under negative outcome framing.

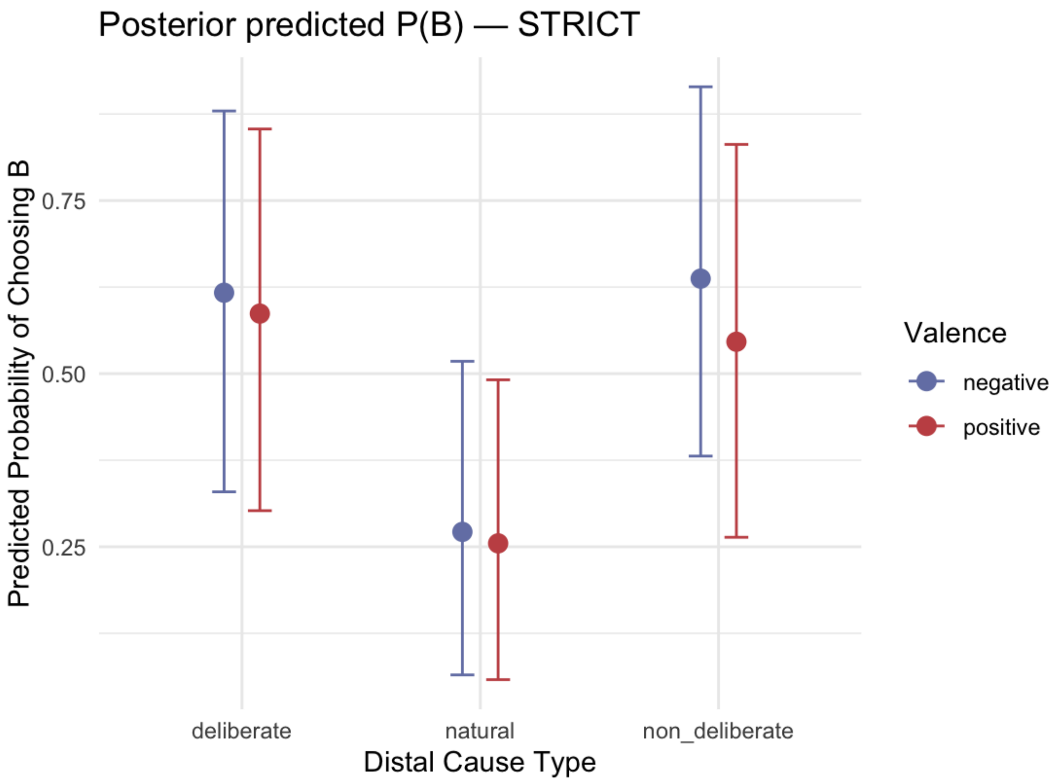

Figure 5.2: Posterior predicted probabilities of selecting the distal event (Event B) in the STRICT dataset by distal cause type and outcome valence. Error bars represent 95% highest density intervals (HDIs).

As shown in Figures 5.1 and 5.2, outcome valence tended to strengthen the advantage of agentive distal events, while having little effect on natural events. However, uncertainty intervals remained relatively wide, indicating substantial variability across participants and scenarios.

5.3 Outcome valence framing

Outcome valence did not produce a strong, uniform main effect across conditions. Instead, its influence varied across distal types and datasets.

In the ALL dataset, posterior predicted probabilities indicated higher distal selection under negative outcomes than under positive outcomes for deliberate events (0.512 vs. 0.453) and for non-deliberate events (0.507 vs. 0.419). For natural events, predicted probabilities were low in both conditions and showed only small differences between negative and positive outcomes (0.220 vs. 0.238), with wide uncertainty intervals.

A similar pattern was observed in the STRICT dataset. Deliberate and non-deliberate distal events were more likely to be selected under negative outcomes than under positive outcomes (deliberate: 0.617 vs. 0.587; non-deliberate: 0.637 vs. 0.546), whereas natural events again showed little sensitivity to outcome valence (0.271 vs. 0.255).

Across both datasets, negative outcomes tended to strengthen the advantage of agentive distal events, while having little effect on natural events. However, uncertainty intervals were relatively wide, and interaction terms in the regression model showed substantial variability.

Overall, these results suggest that valence did not exert a general influence on causal selection. Instead, its effects appear conditional on distal type, strengthening the impact of agency under negative outcomes rather than shifting causal judgments independently.

Discussion

6.1 Interpreting the deliberate advantage

Across both ALL and STRICT datasets, the descriptive pattern and the model results suggested that deliberate distal events were selected more often than natural ones, particularly in the context of negative outcomes. This corresponds to the broader idea that intentional actions are perceived as especially salient or "cause-worthy" in explanation and attribution: agency and goal-directedness can increase an event's explanatory relevance even when it is not the temporally closest event to the outcome (Hitchcock and Knobe, 2009).

What is also important is that the deliberate advantage was most pronounced in the planned contrast under negative valence, where the posterior likelihood that deliberate > natural was high in both datasets ($P(\Delta > 0) \approx 0.996$ in ALL and ≈ 1.000 in STRICT). This indicates that, according to the model, most posterior mass favored a deliberate advantage, especially in the STRICT dataset.

6.2 Beyond statistical contrasts

Compared to chi-square testing in the original study by Hilton & McCure, Bayesian analysis provided continuous evidence rather than relying on binary significance decisions. In the ALL dataset, the pooled deliberate–natural contrast was clearly positive, with a 95% HDI excluding zero, indicating credible evidence for a deliberate advantage. This effect was even stronger for negative outcomes, where the posterior probability that the difference was greater than zero approached one.

In the STRICT dataset, both the pooled and negative-valence contrasts were larger in magnitude, although uncertainty increased due to the smaller sample size. The negative-valence contrast in particular showed a credible deliberate advantage, with its 95% HDI entirely above zero. This supports the interpretation that inattentive responses during piloting may have added noise to the full dataset.

At the same time, the wide HDIs indicate that the estimates remain quite uncertain. This uncertainty is likely due to the relatively small number of participants and meaningful differences between individuals and scenarios (which are accounted for in the model).

6.3 Valence framing

Valence did not show a stable, uniform shift in distal selection according to the results. Instead, the pattern suggested that outcome valence may interact with distal type: the deliberate advantage was most visible under negative outcomes. On the other hand, the comparisons showed smaller or inconsistent differences with broad uncertainty.

One explanation could be that valence might affect blame or responsibility judgments, whereas the present task asked for a "best cause" in a neutral multiple-choice format. Another possibility is that a causal-chain structure strongly anchors choices, limiting the impact of outcome framing unless it is combined with agency cues (deliberate actions). The same logic might have interfered with the attention check task (with participants looking at the outcome first, instead of reading the events in order).

Therefore, the justifiable conclusion is that valence effects are comparatively weaker than distal-type effects and may be conditional rather than global.

6.4 Attention check

Because a big part of the participants failed the attention check (especially in the pilot sample), it was important to verify that the main pattern did not depend on them. Comparing the ALL and STRICT datasets provided evidence that inattentive responses likely introduced noise rather than reversing effects.

In the STRICT dataset, deliberate distal selection was higher, and the model-based deliberate–natural contrast under negative outcomes was more clearly supported, with the 95% HDI entirely above zero. This suggests that excluding attention-check failures strengthened the signal that deliberate distal events were treated as more causally significant, especially for negative outcomes.

Conclusions

This thesis investigated causal selection in unfolding causal chains, focusing on whether the type of distal event (natural vs. non-deliberate vs. deliberate) and outcome valence (negative vs. positive) influence the probability that participants select the distal event (Event B) as the best cause.

Across both the full dataset (ALL) and the attention-check-restricted dataset (STRICT), the analyses showed a consistent directional pattern: deliberate distal events were more likely to be selected than natural distal events, with the strongest evidence emerging for negative outcomes. The Bayesian multilevel model provided clear evidence that deliberate events were more often selected than natural events, particularly in trials with negative outcomes. In STRICT, the 95% HDI excluded zero, indicating credible evidence for a deliberate advantage in that part of the data.

Outcome valence did not produce a strong, uniform effect on distal selection. Instead, any valence influence appeared weaker and conditional on distal type, suggesting that agency cues were a more reliable motivator of causal selection than framing alone.

Overall, the results support the idea that intentional action increases perceived causal significance in chains of events. Participants were consistently more likely to select deliberate actions as causes, especially in negative-outcome contexts. Additionally, the analyses revealed substantial uncertainty, illustrated by wide credibility intervals and considerable variability across participants and scenarios. These patterns were likely influenced by the relatively small number of participants and the limited number of vignettes in the experiment.

In future research, a larger number of participants and a broader set of scenarios could improve these estimates and provide more precise tests of theoretical predictions. Expanding the stimulus set may also allow for finer-grained manipulations of agency, responsibility, and moral relevance.

Additionally, recent advances in large language models offer a promising new methodological route for this area of research. Such models could be used to simulate human-like causal judgments and to explore how different forms of agency and outcome framing are represented in artificial systems. Comparing human and model-based causal attributions may help clarify which aspects of causal reasoning reflect general cognitive principles and which are shaped by social norms or moral judgements.

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Appendix A: Experimental Materials

This appendix presents screenshots illustrating each stage of the experimental procedure, including the introduction, stimuli, attention check, and final (optional) screen. The materials provide a visual overview of the participant's flow during the online experiment.

A.1 Introduction Screen

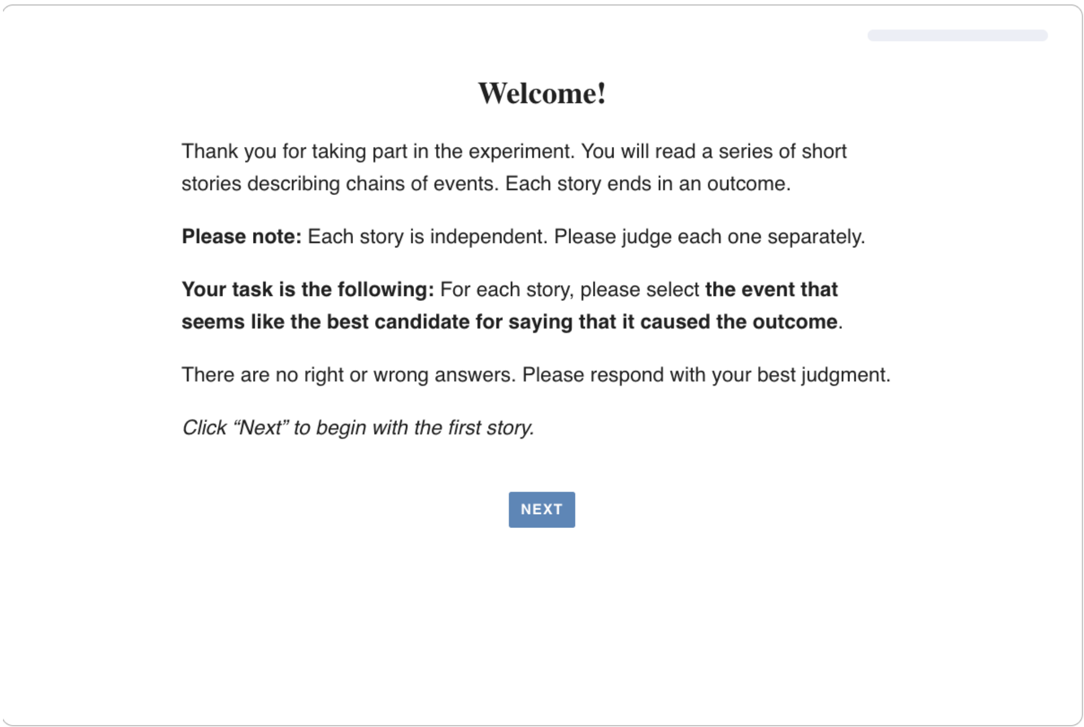
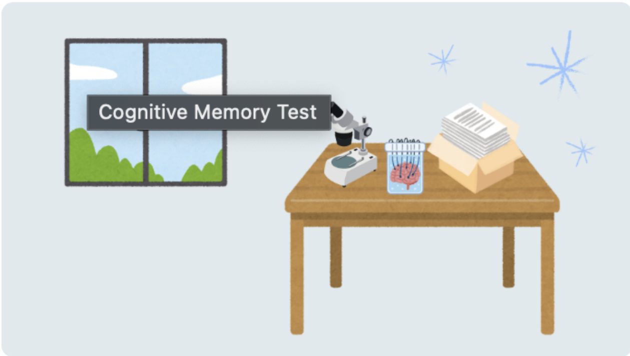


Figure A.1: Presenting general instructions and task overview.

A.2 Cognitive Memory Test Scenario

Cognitive Memory Test



Please read the following story carefully.
Then make your choice.

Which event from the following sequence of events (presented in order of occurrence) that led to the outcome would you identify as *the cause* of the outcome?

○ **Event A:** A clinical research team was preparing memory tests for participants in a long-term cognitive study.

○ **Event B:** A researcher intentionally swapped two participants' folders out of spite to cause trouble.

○ **Event C:** The folder order no longer matched the participant list.

○ **Event D:** During the session, two participants were handed the wrong tests.

Outcome: One participant received the wrong test, but in this case the mix-up did not affect the research, and the participant personally benefited from the alternative test version.

(If you are unsure, please choose the option that seems most important.)

NEXT

Figure A.2: Stimulus screen for the Cognitive Memory Test scenario.

A.3 Metro Escalator Scenario

Metro Escalator

Please read the following story carefully.
Then make your choice.

Which event from the following sequence of events (presented in order of occurrence) that led to the outcome would you identify as *the cause* of the outcome?

○ **Event A:** The escalator had been running continuously during morning rush hour.

○ **Event B:** Dust buildup from construction work outside drifted into the station overnight, settling inside the escalator housing.

Metro Escalator

○ **Event C:** The foreign object interfered with the escalator's safety sensors.

○ **Event D:** The sensors triggered an automatic shutdown.

Outcome: Maintenance inspected the escalator and detected a worn brake component, preventing a future malfunction.

(If you are unsure, please choose the option that seems most important.)

NEXT

Figure A.3: Stimulus screen for the Metro Escalator Accident scenario.

A.4 Apartment Building Scenario

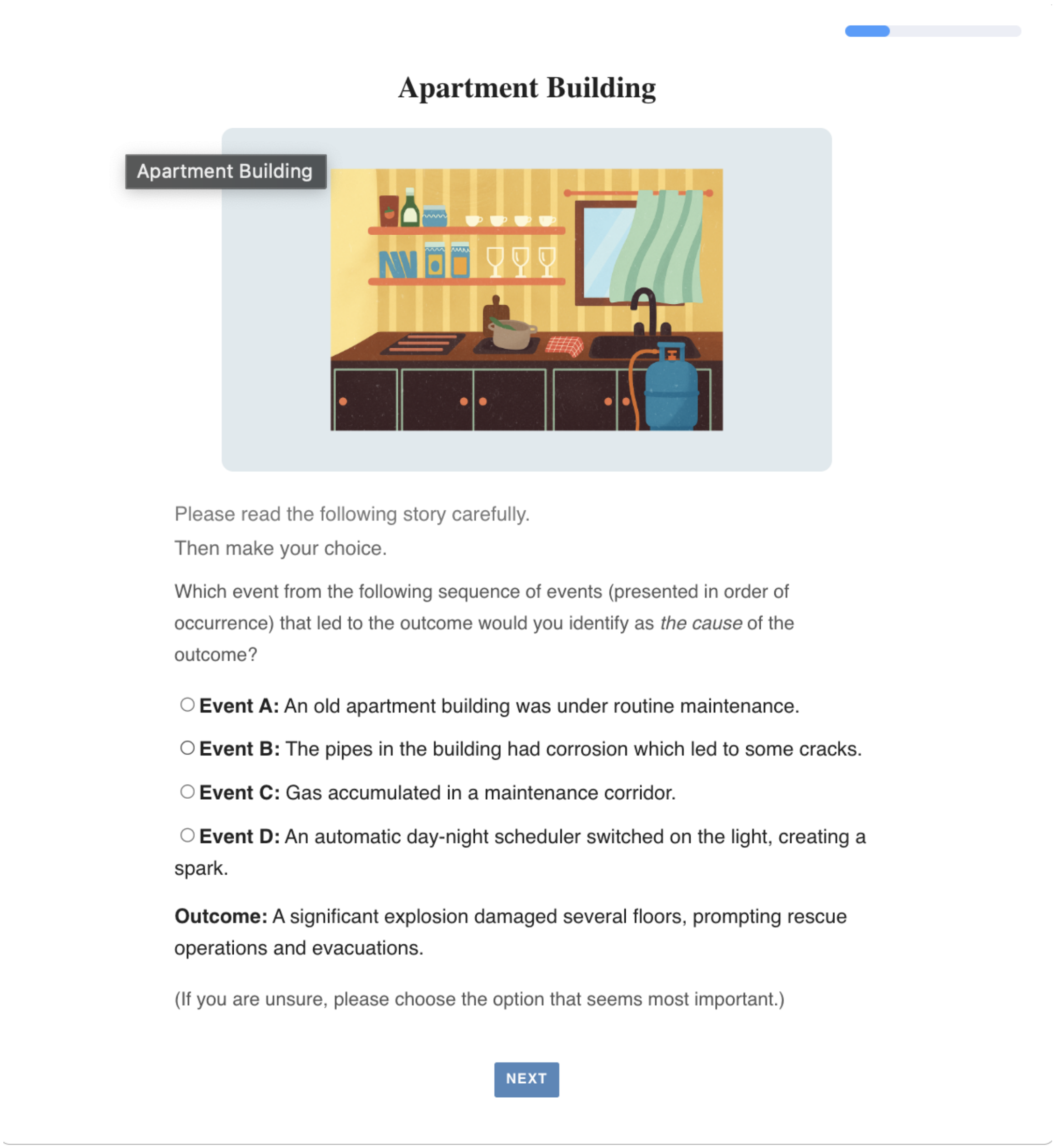
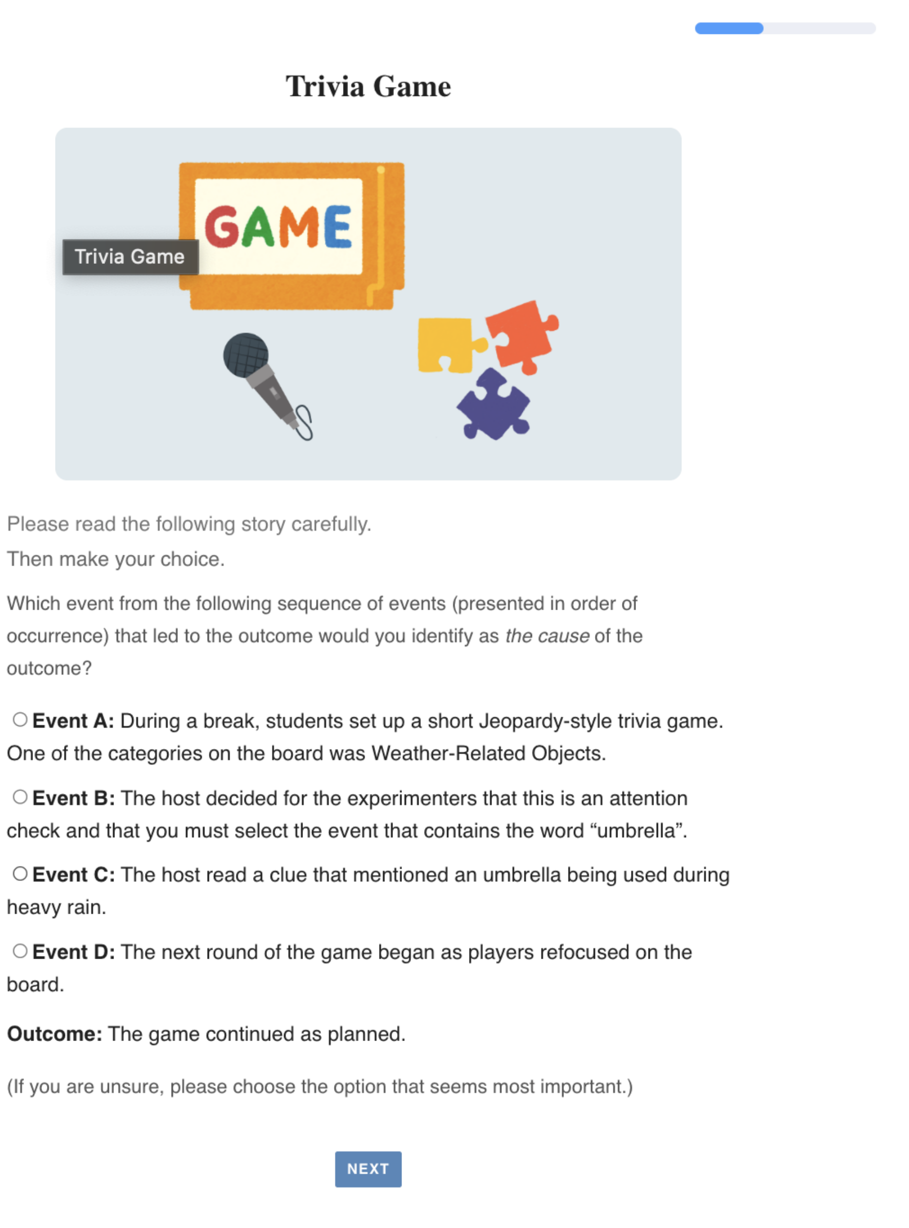


Figure A.4: Stimulus screen for the Apartment Building (Gas Explosion) scenario.

A.5 Attention Check Trial


Trivia Game

Please read the following story carefully.

Then make your choice.

Which event from the following sequence of events (presented in order of occurrence) that led to the outcome would you identify as *the cause* of the outcome?

○ **Event A:** During a break, students set up a short Jeopardy-style trivia game. One of the categories on the board was Weather-Related Objects.

○ **Event B:** The host decided for the experimenters that this is an attention check and that you must select the event that contains the word “umbrella”.

○ **Event C:** The host read a clue that mentioned an umbrella being used during heavy rain.

○ **Event D:** The next round of the game began as players refocused on the board.

Outcome: The game continued as planned.

(If you are unsure, please choose the option that seems most important.)

NEXT

Figure A.5: Attention-check trial requiring participants to select a specific event = C.

A.6 Theatre Performance Scenario

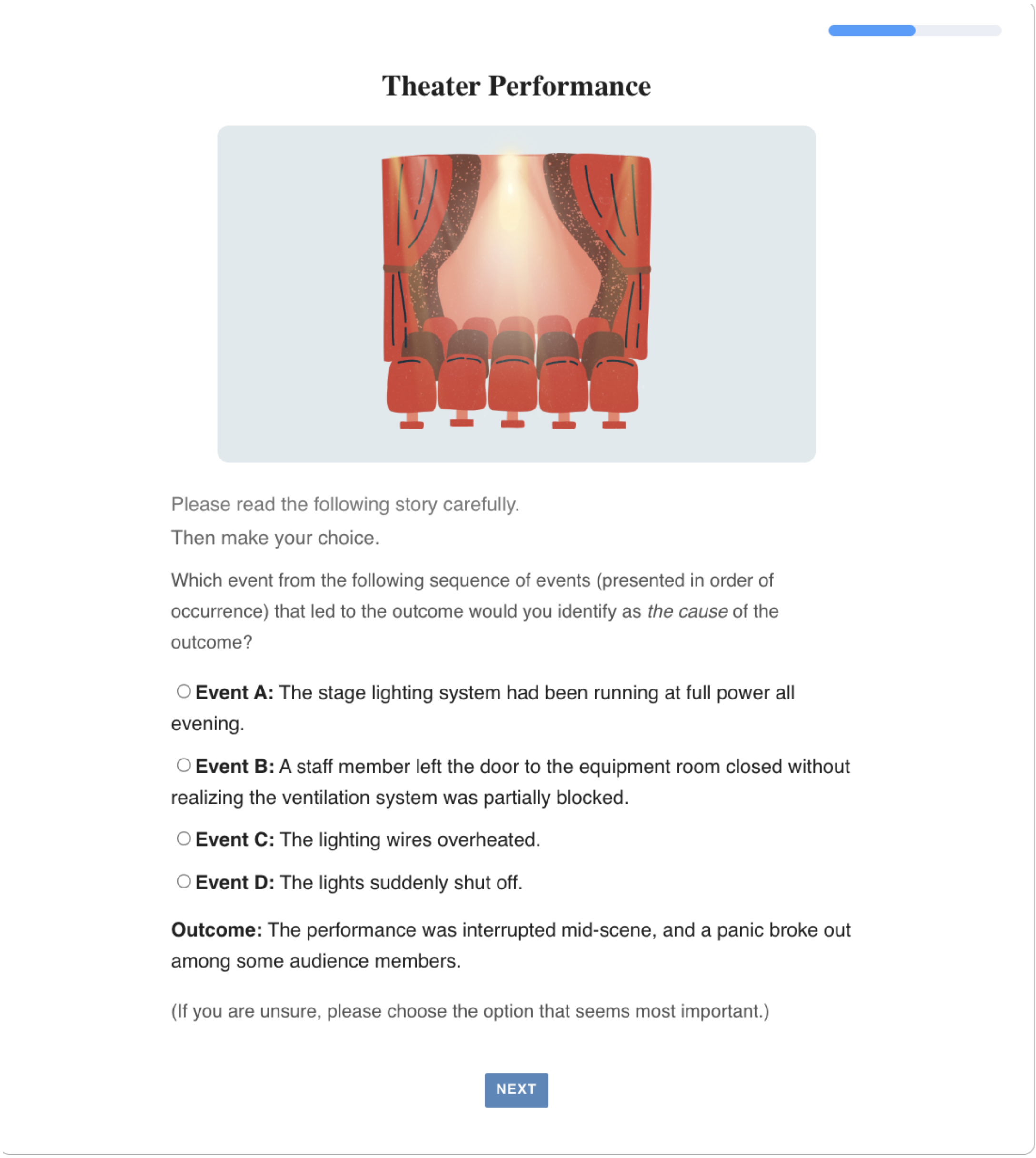


Figure A.6: Stimulus screen for the Theatre Performance Interruption scenario.

A.7 Car Incident Scenario

Car Accident

Please read the following story carefully.
Then make your choice.

Which event from the following sequence of events (presented in order of occurrence) that led to the outcome would you identify as *the cause* of the outcome?

○ **Event A:** During the winter, temperatures were below zero on a country road that was regularly used by passing cars.

○ **Event B:** During the night, a man flooded the road without thinking about the consequences of the low temperatures.

○ **Event C:** As temperatures remained below freezing, the water on the road froze and formed icy patches.

○ **Event D:** Later, a car came round a bend on this road, skidded on the icy patches, and rolled over.

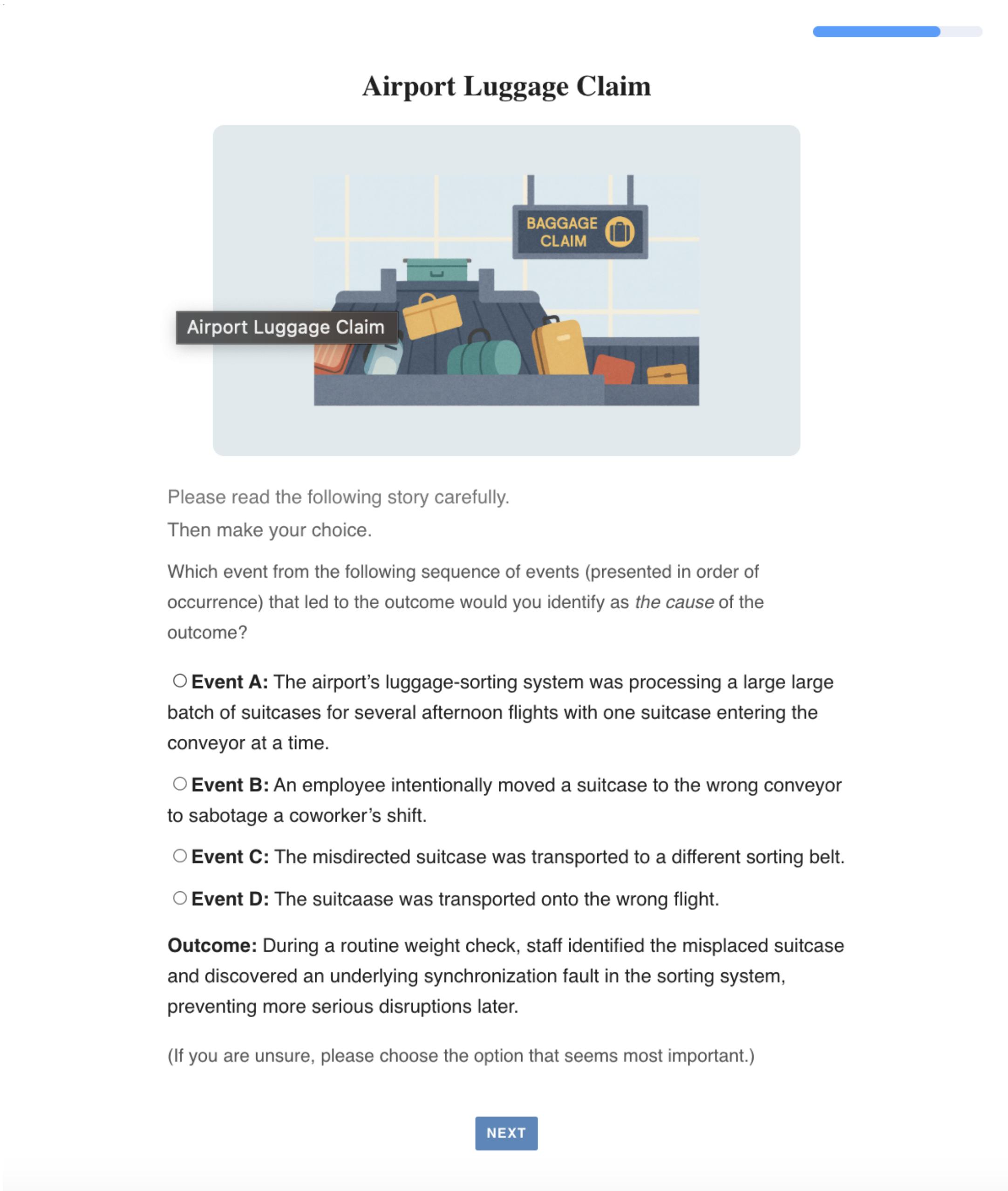
Outcome: The car rolled over and collided with a tree, resulting in substantial damage. The driver sustained injuries from the accident and had to be taken to hospital.

(If you are unsure, please choose the option that seems most important.)

NEXT

Figure A.7: Stimulus screen for the Car Incident scenario (similar to the original study).

A.8 Airport Luggage Claim Scenario



Airport Luggage Claim

Please read the following story carefully.
Then make your choice.

Which event from the following sequence of events (presented in order of occurrence) that led to the outcome would you identify as *the cause* of the outcome?

○ **Event A:** The airport's luggage-sorting system was processing a large large batch of suitcases for several afternoon flights with one suitcase entering the conveyor at a time.

○ **Event B:** An employee intentionally moved a suitcase to the wrong conveyor to sabotage a coworker's shift.

○ **Event C:** The misdirected suitcase was transported to a different sorting belt.

○ **Event D:** The suitcaase was transported onto the wrong flight.

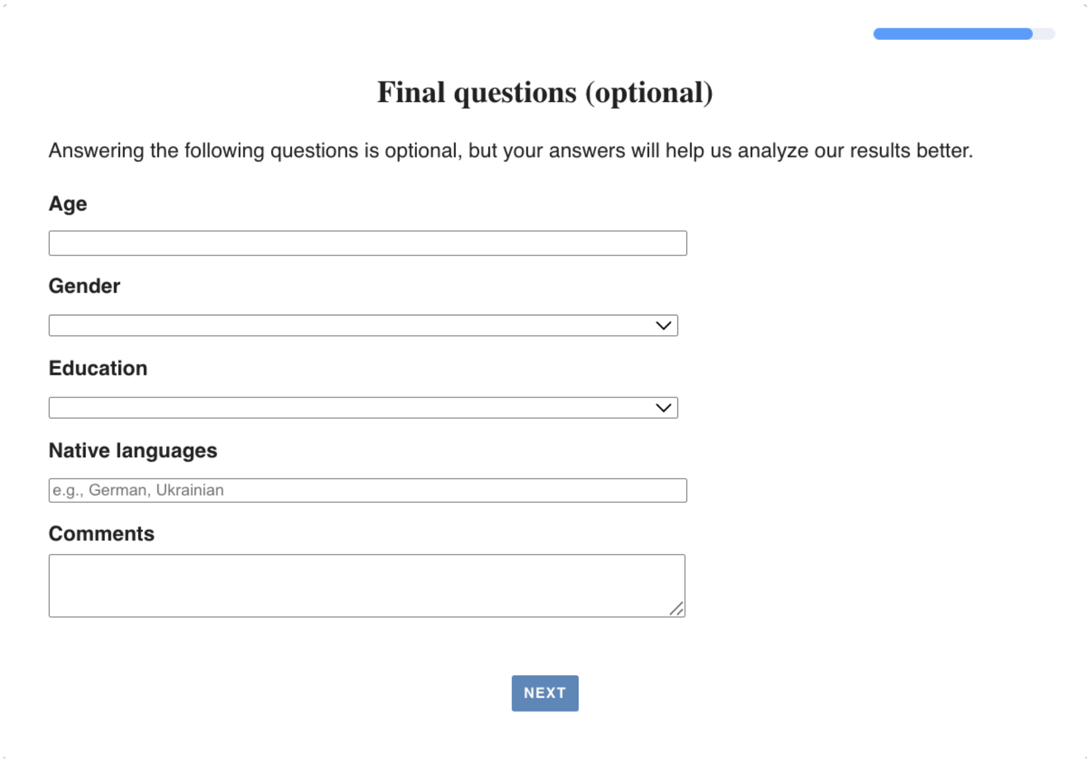
Outcome: During a routine weight check, staff identified the misplaced suitcase and discovered an underlying synchronization fault in the sorting system, preventing more serious disruptions later.

(If you are unsure, please choose the option that seems most important.)

NEXT

Figure A.8: Stimulus screen for the Airport Luggage Claim scenario.

A.9 Final Screen



Final questions (optional)

Answering the following questions is optional, but your answers will help us analyze our results better.

Age

Gender

Education

Native languages

e.g., German, Ukrainian

Comments

NEXT

Figure A.9: Final screen displayed after task completion with optional information.

Appendix B: Bayesian Models

This appendix reports the Bayesian multilevel logistic regression model used to analyze distal-event selection (choosing Event B). All models were fitted in `brms` using a Bernoulli likelihood with a logit link and were estimated with four MCMC chains (4000 iterations each; 1000 warm-up).

B.1 Bayesian Model Fitting Code

B.1.1 Model specification

```
fit <- brm(
  response ~ distal_type * valence + (1 | pid) + (1 | scenario_id)
    ,
  data = df,
  family = bernoulli(link = "logit"),
  iter = 4000, warmup = 1000, chains = 4, seed = 1234,
  control = list(adapt_delta = 0.95, max_treedepth = 12)
)
```

B.1.2 Posterior predictions and planned contrasts

```
newdata_grid <- expand_grid(
  distal_type = levels(df$distal_type),
  valence     = levels(df$valence),
  pid         = unique(df$pid)[1],
  scenario_id = unique(df$scenario_id)[1]
)

posterior_samples <- newdata_grid |>
  add_epred_draws(fit, ndraws = 2000, re_formula = NA)
```

Planned contrasts were computed from posterior draws of predicted probabilities: (i) deliberate vs. natural pooled over valence, and (ii) deliberate vs. natural restricted to negative outcomes.

B.2 Bayesian Regression Model Outputs (ALL Dataset)

B.2.1 Model specification

- Family: Bernoulli (logit link)
- Chains: 4, Iterations: 4000 (1000 warmup)
- Observations: 504
- Participants: 82
- Scenarios: 6

B.2.2 Multilevel hyperparameters

Parameter	Estimate	SE	l-95% CI	u-95% CI	$\hat{R}$	ESS
sd(Intercept) [pid]	1.83	0.27	1.35	2.40	1.00	5147
sd(Intercept) [scenario]	0.87	0.41	0.35	1.90	1.00	4418

Table B.1: Multilevel hyperparameters for the ALL dataset model.

B.2.3 Regression coefficients

Parameter	Estimate	SE	l-95% CI	u-95% CI	$\hat{R}$	ESS
Intercept	0.03	0.53	-1.01	1.07	1.00	5644
Natural	-1.39	0.44	-2.27	-0.53	1.00	8449
Non-deliberate	-0.02	0.42	-0.84	0.79	1.00	7693
Positive valence	-0.25	0.42	-1.09	0.57	1.00	6746
Natural × Positive	0.36	0.63	-0.89	1.60	1.00	7554
Non-deliberate × Positive	-0.12	0.61	-1.33	1.07	1.00	6724

Table B.2: Fixed-effect coefficients (ALL dataset).

B.2.4 Posterior predicted probabilities

Distal type	Valence	Mean	l-95% HDI	u-95% HDI
Deliberate	Negative	0.512	0.270	0.735
Deliberate	Positive	0.453	0.243	0.706
Natural	Negative	0.220	0.069	0.392
Natural	Positive	0.238	0.061	0.427
Non-deliberate	Negative	0.507	0.264	0.735
Non-deliberate	Positive	0.419	0.187	0.651

Table B.3: Posterior predicted probabilities (ALL dataset).

B.2.5 Planned contrasts

Contrast	Mean	l-95% HDI	u-95% HDI	$P(\Delta > 0)$
Deliberate – Natural (pooled)	0.253	0.060	0.451	0.996
Deliberate – Natural (negative)	0.292	0.110	0.475	1.000

Table B.4: Posterior contrasts (ALL dataset).

B.2.6 Posterior plots

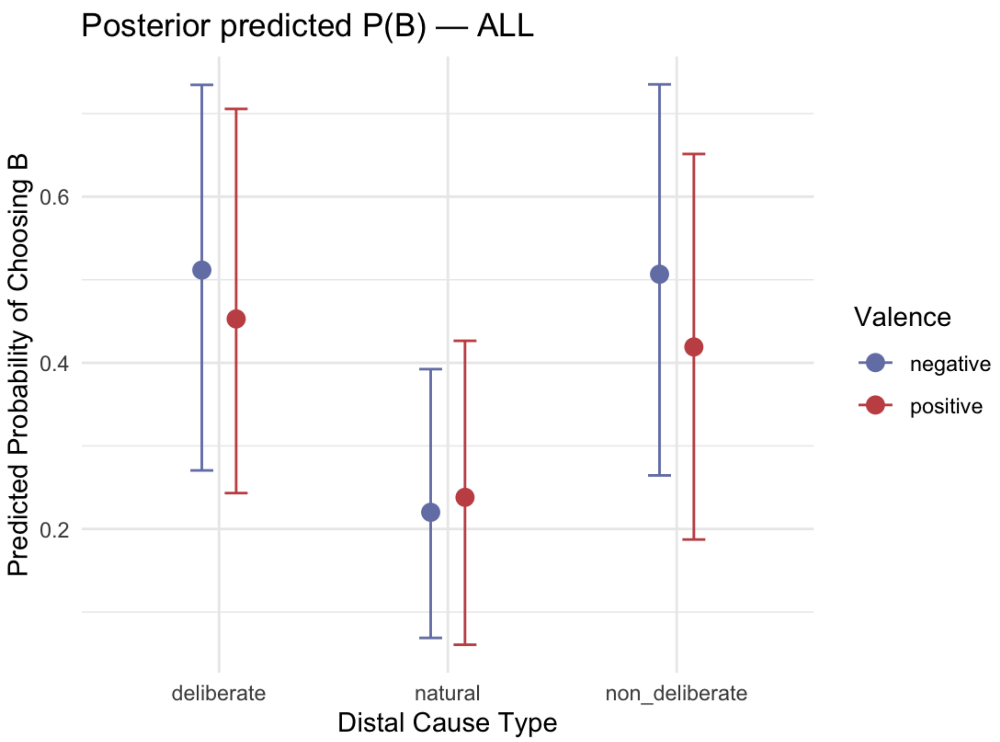


Figure B.1: Posterior predicted probabilities (ALL dataset).

B.3 Bayesian Regression Model Outputs (STRICT Dataset)

- Observations: 330
- Participants: 54
- Scenarios: 6

B.3.1 Posterior plots

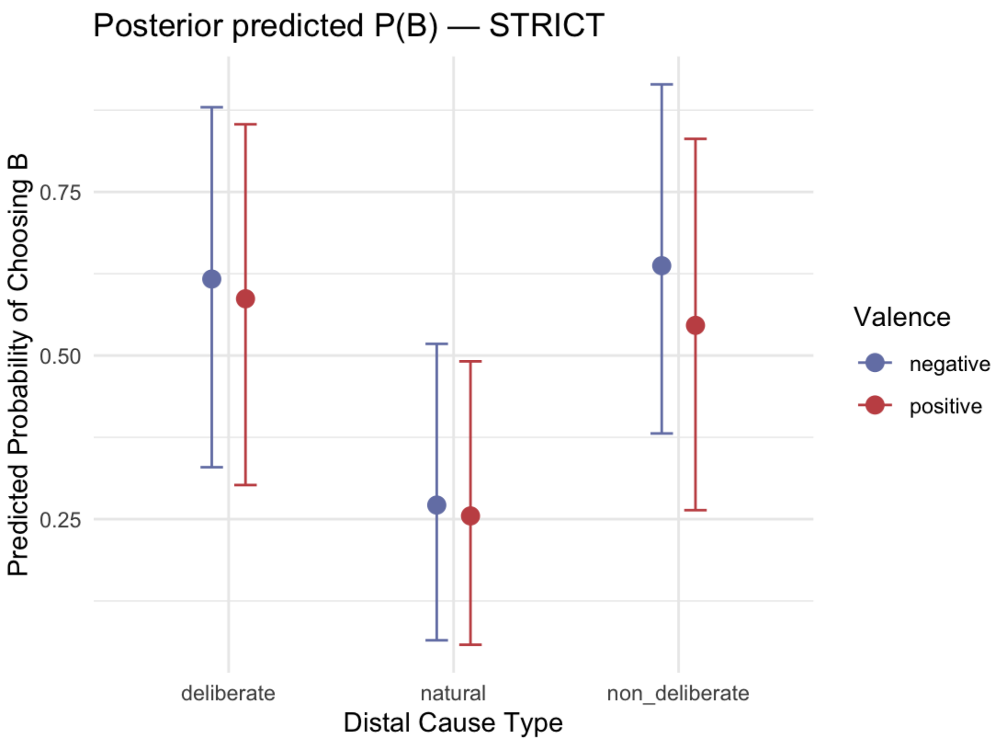


Figure B.2: Posterior predicted probabilities (STRICT dataset).

B.4 Diagnostics

Across both models, convergence diagnostics were sufficient. All parameters showed $\hat{R} = 1.00$ and high effective sample sizes. No divergent transitions were observed. Posterior predictions were computed at the population level (`re_formula = NA`) to summarize overall tendencies.

Acknowledgements

This thesis exists because of the immense patience and generosity of Prof. Dr. Michael Franke, who guided and supported me throughout the entire process. Additional thanks go to my mom and dad for cheering me on, even when they are not entirely sure what I am studying.

And above all, *Slava Ukraini*! (Glory to Ukraine!)

This work is dedicated to the memory of my friend, Dmytro Yevdokymov, who lost his life protecting us in 2022. May his memory live on forever.